

Assignment no. 2

Natural Language Processing
NLP4IL (AIS.ba VZ)
WS 2026

This assignment sheet covers the augmentation and tokenization of text. Use the following five sentences as content for all following exercises.

1. Production on the film began in June 1921 with Colleen's return from New York and Florida, where she was making *The Lotus Eater* with John Barrymore which was directed by Marshall Neilan.
2. The device operates as long as the magic smoke is trapped inside of it, but when the smoke escapes from it, the device ceases to operate.
3. The 1980s farm crisis caused some families to have to give up their farms, and farms have been merged to industrial scale.
4. From conceptual structuration, the architecture on which the co-simulation framework is developed and the formal semantic relations/syntactic formulation are defined.
5. The chest is a chestnut colour with thin black bars, while long black-margined plumes arise from its flanks.

Exercise 1 Simple Data Augmentation

Implement a simple data augmentation technique by randomly swapping words. Create an algorithm that swaps a specified ratio of words in the sentence. Make sure to swap the same word just once.

Task

- Generate 5 augmented sentences from the input corpus using random swaps.

Exercise 2 Synonym Replacement Augmentation Use WordNet from [NLTK](#) to replace words with synonyms. Make the ratio of word replacements configurable.

Tasks

- Retrieve synonyms for selected words
- Replace words in a sentence with synonyms
- Generate at least 3 augmented versions of the input corpus

Exercise 3 Back-Translation Augmentation

Use the [transformers](#) package to augment data by back-translation. The following provides a function to use transformers to translate. There are many models on HuggingFace, one possible option is Helsinki-NLP/opus-mt_tiny_eng-fra.

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM, AutoConfig

def roundtrip_translate(sentences, model_a_name, model_b_name):
    # Load models + tokenizers
    tokenizer_a = AutoTokenizer.from_pretrained(model_a_name)
    model_a = AutoModelForSeq2SeqLM.from_pretrained(model_a_name)

    tokenizer_b = AutoTokenizer.from_pretrained(model_b_name)
    model_b = AutoModelForSeq2SeqLM.from_pretrained(model_b_name)

    results = []

    for text in sentences:
        inputs_a = tokenizer_a(text, return_tensors="pt")
        outputs_a = model_a.generate(**inputs_a)
        translated = tokenizer_a.decode(outputs_a[0], skip_special_tokens=True)

        inputs_b = tokenizer_b(translated, return_tensors="pt")
        outputs_b = model_b.generate(**inputs_b)
        back_translated = tokenizer_b.decode(outputs_b[0], skip_special_tokens=True)

        results.append((translated, back_translated))

    return results
```

Tasks

- Augment the input corpus by backtranslation.
- Generate at least 3 augmented versions by using different languages.

Exercise 4 Basic Tokenization

Write a Python script that performs word and sentence tokenization on the input corpus using NLTK.

- Install the required libraries (nltk).
- Download tokenizer resources using `nltk.download()`.
- Tokenize the text

Tasks

- Perform sentence tokenization
- Perform word tokenization
- Print the resulting tokens of the input corpus

Exercise 5 Tokenization with spaCy

Use [spaCy](#) to tokenize text and analyze linguistic features.

```
import spacy
nlp = spacy.load("en_core_web_sm")

text = "Text to tokenize."
doc = nlp(text)
```

Tasks

- Print all tokens
- Print token POS tags
- Print token lemmas

When using Large Language Models (eg. ChatGPT, Copilot, ...) to produce output that is handed in, also state the prompt and model used in the submission. Submissions that are not marked and are noticeably AI generated will be graded with zero points!